제어공학2
Automatic Control

State-Space Analysis and Controller Design

CHANGWON NATIONAL UNIVERSITY

CHANGWON NATIONAL UNIVERSITY

상태방정식 State equations

- ✓전달함수 방법에 대한 대안
- ✓고계(higher-order) 미분방정식을 분석하기 위한 강력한 수단
- ✓현대 제어이론에 많이 사용

$$M\frac{d^2y(t)}{dt^2}(t)+B\frac{dy(t)}{dt}+Ky(t)=f(t) \quad \Rightarrow \quad \frac{dx_2(t)}{dt}=-\frac{B}{M}x_2(t)-\frac{K}{M}x_1(t)+\frac{1}{M}f(t)$$

$$x_1(t)=y(t) \qquad x_2(t)=\frac{dx_1(t)}{dt}=\frac{dy(t)}{dt}$$

$$\frac{dx_1(t)}{dt}=x_2(t)$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

✓상태 방정식 State equations

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

✓상태 벡터 State vector

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

✓입력 벡터 Input vector

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_p(t) \end{bmatrix}$$

✓계수 행렬 Coefficient matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} (n \times n)$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} (n \times p)$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

✓출력 방정식 Output equation

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_q(t) \end{bmatrix} = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{q1} & c_{q2} & \cdots & c_{qn} \end{bmatrix} \qquad \mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1p} \\ d_{21} & d_{22} & \cdots & d_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ d_{q1} & d_{q2} & \cdots & d_{qp} \end{bmatrix}$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

- ✓상태 방정식 State equations
- ✓출력 방정식 Output equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 2u(t)$$

$$y(t) = x_1(t)$$

$$x_1(t) = y(t) \qquad \frac{dx_1(t)}{dt} = x_2(t)$$

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}; \quad \mathbf{u}(t) = u(t)$$

$$x_2(t) = \frac{dx_1(t)}{dt} = \frac{dy(t)}{dt} \qquad \frac{dx_2(t)}{dt} = -2x_1(t) - 3x_2(t) + 2u(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}; \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}; \quad \mathbf{D} = 0$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

✓라플라스 변환 Laplace transform

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(t) \qquad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix};\ \mathbf{B} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s)]$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2 + 3s + 2}\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$\mathbf{X}(s) = L[\mathbf{x}(t)] \text{ and } \mathbf{U}(s) = L[\mathbf{u}(t)]$$

$$\mathbf{x}(t) == \mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}]\mathbf{x}(0) + \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s)]\}$$

$$\mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}]\mathbf{B}\mathbf{U}(s) = \mathcal{L}^{-1}\left(\frac{1}{s^2 + 3s + 2}\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix}\frac{1}{s}\right)$$

$$= \mathcal{L}^{-1}\left(\frac{1}{s^2 + 3s + 2}\begin{bmatrix} \frac{2}{s} \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 - e^{-t} + e^{-2t} \\ e^{-t} - 2e^{-2t} \end{bmatrix} \quad t \geq 0$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

✓전달함수 Transfer functions

$$\frac{d^n y(t)}{dt^n} = a_{n-1}\frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t)$$

$$= b_m \frac{d^m u(t)}{dt^m} + b_{m-1}\frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

$$G_{ij}(s) = \frac{Y_i(s)}{U_j(s)}$$

$$Y_i(s) = G_{i1}(s)U_1(s) + G_{i2}(s)U_2(s) + \cdots + G_{ip}(s)U_p(s)$$

$$\mathbf{G}(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \cdots & G_{1p}(s) \\ G_{21}(s) & G_{22}(s) & \cdots & G_{2p}(s) \\ . & . & \cdots & . \\ G_{q1}(s) & G_{q2}(s) & \cdots & G_{qp}(s) \end{bmatrix}$$

CHANGWON NATIONAL UNIVERSITY

상태공간 형태 State space form

✓특성 방정식 Characteristic equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$|s\mathbf{I} - \mathbf{A}| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 2 & 1 & s+5 \end{vmatrix} = s^3 + 5s^2 + s + 2 = 0$$

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} = \mathbf{C}\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\mathbf{B} + \mathbf{D}$$

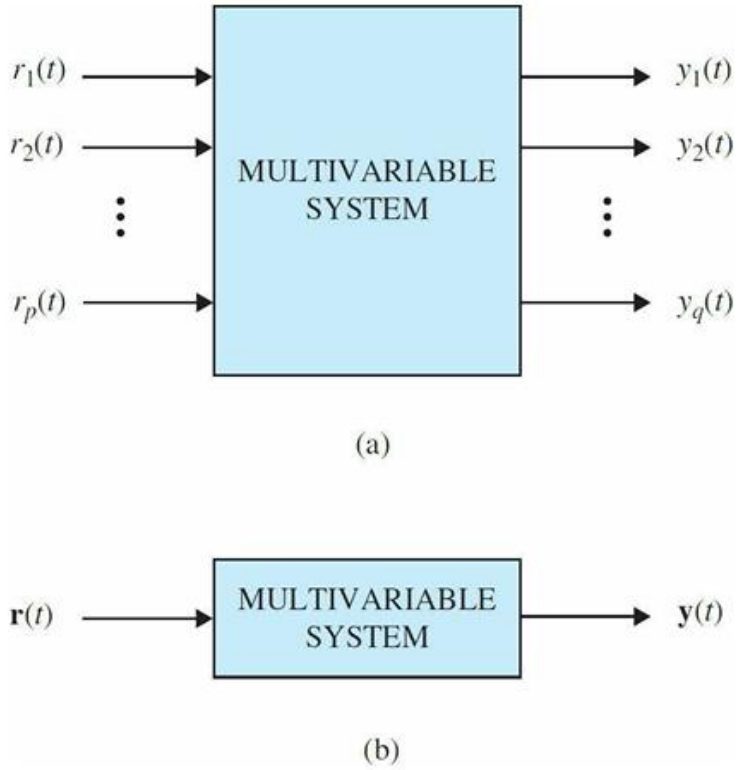
$$= \frac{\mathbf{C}[\text{adj}(s\mathbf{I} - \mathbf{A})]\mathbf{B} + |s\mathbf{I} - \mathbf{A}|\mathbf{D}}{|s\mathbf{I} - \mathbf{A}|}$$

$$|s\mathbf{I} - \mathbf{A}| = \begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = s^2 + 3s + 2 = 0$$

$$|s\mathbf{I} - \mathbf{A}| = 0$$

블록선도, 전달함수, 상태선도

✓전달함수 (다변수시스템) Transfer function (Multivariable systems)



$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{R}(s)$$

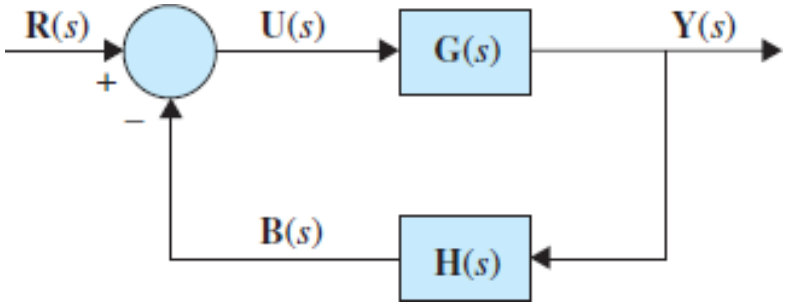
$$\mathbf{Y}(s) = \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_q(s) \end{bmatrix} \qquad \mathbf{G}(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \cdots & G_{1p}(s) \\ G_{21}(s) & G_{22}(s) & \cdots & G_{2p}(s) \\ . & . & \cdots & . \\ G_{q1}(s) & G_{q2}(s) & \cdots & G_{qp}(s) \end{bmatrix}$$

$$\mathbf{R}(s) = \begin{bmatrix} R_1(s) \\ R_2(s) \\ \vdots \\ R_p(s) \end{bmatrix}$$

Figure 8-1 Block diagram representations of a multivariable system.

블록선도, 전달함수, 상태선도

✓블록선도 Block diagram



$$\mathbf{Y}(s)=\mathbf{G}(s)\mathbf{U}(s)$$

$$\mathbf{U}(s)=\mathbf{R}(s)-\mathbf{B}(s)$$

$$\mathbf{B}(s)=\mathbf{H}(s)\mathbf{Y}(s)$$

$$\mathbf{Y}(s)=\mathbf{G}(s)\mathbf{R}(s)-\mathbf{G}(s)\mathbf{H}(s)\mathbf{Y}(s)$$

→ **I+G(s)H(s)** is nonsingular

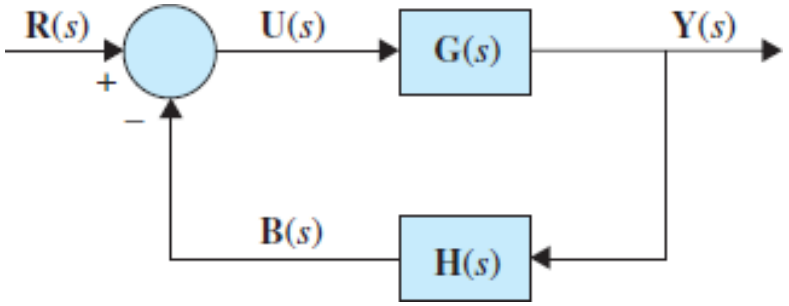
$$\mathbf{Y}(s)=[\mathbf{I}+\mathbf{G}(s)\mathbf{H}(s)]^{-1}\mathbf{G}(s)\mathbf{R}(s)$$

$$\mathbf{M}(s)=[\mathbf{I}+\mathbf{G}(s)\mathbf{H}(s)]^{-1}\mathbf{G}(s)$$

$$\mathbf{Y}(s)=\mathbf{M}(s)\mathbf{R}(s)$$

블록선도, 전달함수, 상태선도

✓블록선도 Block diagram



$$\mathbf{Y}(s)=[\mathbf{I}+\mathbf{G}(s)\mathbf{H}(s)]^{-1}\mathbf{G}(s)\mathbf{R}(s)$$

$$\mathbf{G}(s)=\begin{bmatrix}\frac{1}{s+1} & -\frac{1}{s}\\ 2 & \frac{1}{s+2}\end{bmatrix}\quad \mathbf{H}(s)=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\mathbf{I}+\mathbf{G}(s)\mathbf{H}(s)=\begin{bmatrix}1+\frac{1}{s+1} & -\frac{1}{s}\\ 2 & 1+\frac{1}{s+2}\end{bmatrix}=\begin{bmatrix}\frac{s+2}{s+1} & -\frac{1}{s}\\ 2 & \frac{s+3}{s+2}\end{bmatrix}$$

$$\mathbf{M}(s)=\frac{s(s+1)}{s^2+5s+2}\begin{bmatrix}\frac{3s^2+9s+4}{s(s+1)(s+2)} & -\frac{1}{s}\\ 2 & \frac{3s+2}{s(s+1)}\end{bmatrix}$$

1차 미분방정식 시스템: 상태방정식

✓상태변수의 정의 Definition of state variables

$$x_1(t) = y(t)$$

$$x_2(t) = \frac{dy(t)}{dt}$$

$$\vdots$$

$$x_n(t) = \frac{d^{n-1}y(t)}{dt^{n-1}}$$

$$\frac{dx_1(t)}{dt} = x_2(t)$$

$$\frac{dx_2(t)}{dt} = x_3(t)$$

$$\vdots$$

$$\frac{dx_n(t)}{dt} = -a_0x_1(t) - a_1x_2(t) - \cdots - a_{n-2}x_{n-1}(t) - a_{n-1}x_n(t) + f(t)$$

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

상태방정식의 벡터-행렬 표현

✓상태변수의 정의 Definition of state variables

- 상태벡터 State vector

$$\mathbf{x}(t)=\begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

- 입력벡터 Input vector

$$\mathbf{u}(t)=\begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_p(t) \end{bmatrix}$$

- 계수행렬 Coefficient matrix

$$\mathbf{A}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}(n\times n)$$

$$\mathbf{B}=\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}(n\times p)$$

상태방정식의 벡터-행렬 표현

✓출력 방정식 Output equation

$$\mathbf{y}(t)=\begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_q(t) \end{bmatrix}=\mathbf{Cx}(t)+\mathbf{Du}(t)$$

$$\mathbf{C}=\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{q1} & c_{q2} & \cdots & c_{qn} \end{bmatrix}$$

$$\mathbf{D}=\begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1p} \\ d_{21} & d_{22} & \cdots & d_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ d_{q1} & d_{q2} & \cdots & d_{qp} \end{bmatrix}$$

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=2u(t)$$

$$\frac{dx_1(t)}{dt}=x_2(t)$$

$$\frac{dx_2(t)}{dt}=-2x_1(t)-3x_2(t)+2u(t)$$

$$\mathbf{x}(t)=\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix};\quad \mathbf{u}(t)=u(t)$$

$$\mathbf{A}=\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix};\quad \mathbf{B}=\begin{bmatrix} 0 \\ 2 \end{bmatrix};\quad \mathbf{C}=\begin{bmatrix} 1 & 0 \end{bmatrix};\quad \mathbf{D}=0$$

상태천이행렬 State-transition matrix

- 선형제차 상태방정식을 만족시키는 행렬

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) \qquad \frac{d\phi(t)}{dt} = \mathbf{A}\phi(t)$$

$$\mathbf{x}(t) = \phi(t)\mathbf{x}(0)$$ which is the solution of the homogeneous state equation for t ≥ 0

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s)$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0)$$

$$\mathbf{x}(t) = \mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}]\mathbf{x}(0) \quad t \geq 0$$

$$\phi(t) = \mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}]$$ 상태천이행렬

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$$

$$\frac{de^{\mathbf{A}t}}{dt} = \mathbf{A}e^{\mathbf{A}t}$$

$$\phi(t) = \mathbf{e}^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{A}^2t^2 + \frac{1}{3!}\mathbf{A}^3t^3 + \cdots$$

$X(s) = (sI - A)^{-1}X(0)$

$x(t) = \mathcal{L}^{-1}\{(sI - A)^{-1}\}x(0)$

$\Phi(t)$

$\rightarrow x(t) = \Phi(t)x(0)$

(상태천이행렬

상태천이행렬 State-transition matrix

✓상태천이행렬

- 초기조건에 의한 응답 — 추가 입력값이 없는 상태일때
- A의 상태천이행렬
- 입력이 0일때 $t = 0$인 초기시간으로부터 임의의 t 까지의 상태천이를 정의

✓상태천이행렬의 성질 Properties of state-transition matrix

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) \qquad \phi(t) = \mathbf{e}^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{A}^2t^2 + \frac{1}{3!}\mathbf{A}^3t^3 + \cdots$$

$$\phi(0) = \mathrm{I} \quad \text{(the identity matrix)}$$

$$\phi^{-1}(t) = \phi(-t) \quad \rightarrow \quad (e^{At})^{-1} = e^{-At}$$

$$\phi(t_2 - t_1)\phi(t_1 - t_0) = \phi(t_2 - t_0)$$

$$[\phi(t)]^k = \phi(kt) \quad \text{for } k = \text{positive integer}$$

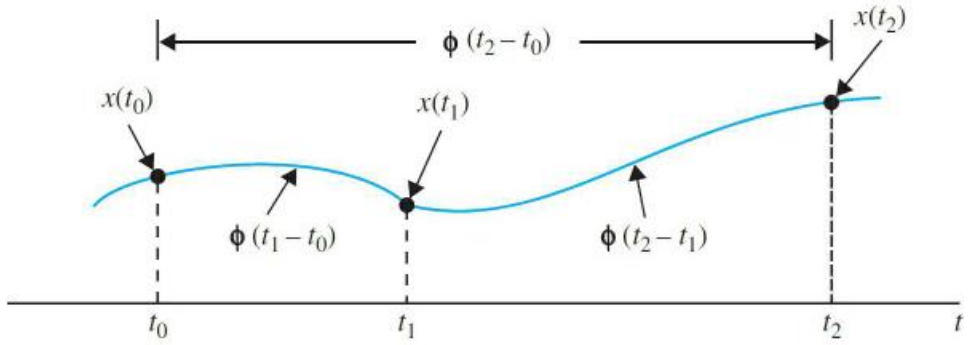


Figure 8-3 Property of the state-transition matrix.

상태천이 방정식 State-transition equation

$$\frac{\mathbf{dx}(t)}{dt}=\mathbf{Ax}(t)+\mathbf{Bu}(t)+\mathbf{Ew}(t)$$

$$s\mathbf{X}(s)-\mathbf{x}(0)=\mathbf{AX}(s)+\mathbf{BU}(s)+\mathbf{EW}(s)$$

$$\mathbf{X}(s)=(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{x}(0)+(s\mathbf{I}-\mathbf{A})^{-1}[\mathbf{BU}(s)+\mathbf{EW}(s)]$$

$$\mathbf{x}(t)=\mathcal{L}^{-1}[(s\mathbf{I}-\mathbf{A})^{-1}]\mathbf{x}(0)+\mathcal{L}^{-1}\{(s\mathbf{I}-\mathbf{A})^{-1}[\mathbf{BU}(s)+\mathbf{EW}(s)]\}$$

$$=\phi(t)\mathbf{x}(0)+\int_0^t\phi(t-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau\quad t\geq 0$$

$$\mathbf{x}(0)=\phi(-t_0)\mathbf{x}(t_0)-\phi(-t_0)\int_0^{t_0}\phi(t_0-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau$$

$$F_1(s)F_2(s)=\mathcal{L}\left[\int_0^t f_1(\tau)f_2(t-\tau)d\tau\right]$$

$$\mathbf{x}(t)=\phi(t)\phi(-t_0)\mathbf{x}(t_0)-\phi(t)\phi(-t_0)\int_0^{t_0}\phi(t_0-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau$$

$$+\int_0^t\phi(t-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau$$

$$\mathbf{x}(t)=\phi(t-t_0)\mathbf{x}(t_0)+\int_0^t\phi(t-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau\quad t\geq t_0$$

$$\mathbf{y}(t)=\mathbf{Cx}(t)+\mathbf{Du}(t)+\mathbf{Hw}(t)$$

$$y(t)=\mathbf{C}\phi(t-t_0)\mathbf{x}(t_0)+\int_{t_0}^t\mathbf{C}\phi(t-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau$$

$$+\mathbf{Du}(t)+\mathbf{Hw}(t)\quad t\geq t_0$$

상태천이 방정식 State-transition equation

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$(s\mathbf{I}-\mathbf{A})^{-1} = \frac{1}{s^2+3s+2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}[(s\mathbf{I}-\mathbf{A})^{-1}] = \begin{bmatrix} 2e^{-t}-e^{-2t} & e^{-t}-e^{-2t} \\ -2e^{-t}+2e^{-2t} & -e^{-t}+2e^{-2t} \end{bmatrix}$$

$$\mathbf{x}(t) = \phi(t-t_0)\mathbf{x}(t_0) + \int_0^t \phi(t-\tau)[\mathbf{B}\mathbf{u}(\tau)+\mathbf{E}\mathbf{w}(\tau)]d\tau \quad t \geq t_0$$

$$\mathbf{x}(t) = \begin{bmatrix} 2e^{-t}-e^{-2t} & e^{-t}-e^{-2t} \\ -2e^{-t}+2e^{-2t} & -e^{-t}+2e^{-2t} \end{bmatrix} x(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)}-e^{-2(t-\tau)} & e^{-(t-\tau)}-e^{-2(t-\tau)} \\ -2e^{-(t-\tau)}+e^{-2(t-\tau)} & -e^{-(t-\tau)}+2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau$$

$$\mathbf{x}(t) = \begin{bmatrix} 2e^{-t}-e^{-2t} & e^{-t}-e^{-2t} \\ -2e^{-t}+2e^{-2t} & -e^{-t}+2e^{-2t} \end{bmatrix} \mathbf{x}(0) + \begin{bmatrix} 0.5-e^{-t}+0.5e^{-2t} \\ e^{-t}-e^{-2t} \end{bmatrix} \quad t \geq 0$$

$$\mathcal{L}^{-1}[(s\mathbf{I}-\mathbf{A})^{-1}]\mathbf{B}\mathbf{U}(s) = {}^{-1}\left(\frac{1}{s^2+3s+2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s} \right)$$

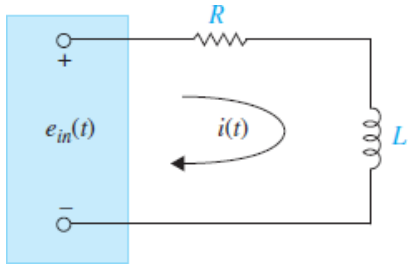
$$= {}^{-1}\left(\frac{1}{s^2+3s+2} \begin{bmatrix} \frac{1}{s} \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0.5-e^{-t}+0.5e^{-2t} \\ e^{-t}-e^{-2t} \end{bmatrix} \quad t \geq 0$$

CHANGWON NATIONAL UNIVERSITY

상태천이 방정식 State-transition equation

$e_{in} = Ri(t) + L\frac{di(t)}{dt} \rightarrow \frac{di(t)}{dt} = -\frac{R}{L}i(t) + \frac{1}{L}e_{in}(t)$ (A, B)

$\phi = e^{At}$, $\phi(t) = e^{-\frac{R}{L}t}$



$$\frac{di(t)}{dt} = -\frac{R}{L}i(t) + \frac{1}{L}e_{in}(t)$$

$$A = -\frac{R}{L} \quad B = \frac{1}{L} \quad E = 0$$

$$\phi(t) = e^{-At} = e^{-Rt/L}$$

$$\mathbf{x}(t) = \mathcal{L}^{-1}[(s\mathbf{I}-\mathbf{A})^{-1}]\mathbf{x}(0) + \mathcal{L}^{-1}\{(s\mathbf{I}-\mathbf{A})^{-1}[\mathbf{BU}(s)+\mathbf{EW}(s)]\}$$

$$= \phi(t)\mathbf{x}(0) + \int_0^t \phi(t-\tau)[\mathbf{Bu}(\tau)+\mathbf{Ew}(\tau)]d\tau \quad t \geq 0$$

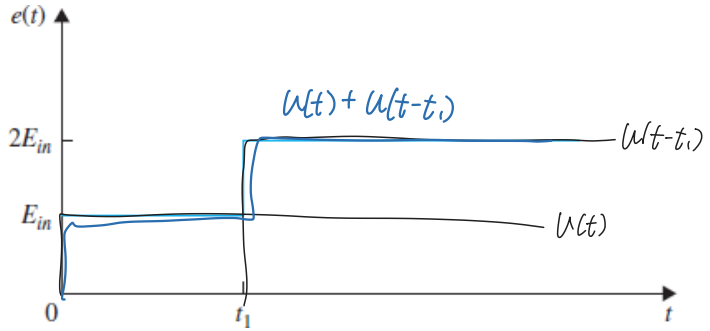
($i(t) =$, $\phi(t)\mathbf{x}(0) = e^{-\frac{R}{L}t}i(0)$)

$$(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{BU}(s) = \frac{E_{in}}{Ls(s+R/L)}(1+e^{-t_1 s})$$

$$i(t) = e^{-Rt/L}i(0)u_s(t) + \frac{E_{in}}{R}(1-e^{-Rt/L})u_s(t) + \frac{E_{in}}{R}(1-e^{-R(t-t_1)/L})u_s(t-t_1)$$

$E_{in} = RI(s) + sLI(s)$

$= (R + sL)I(s)$



$E_{in}[u(t) + u(t-t_1)]$

$\rightarrow \left[\frac{1}{s} + \frac{1}{s}e^{-t_1 s}\right] E_{in}$

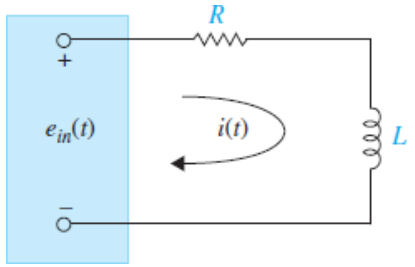
$$e(t) = E_{in}u_s(t) + E_{in}u_s(t-t_1)$$

$\downarrow \mathcal{L}$

$$E(s) = \frac{E_{in}}{s}(1+e^{-t_1 s})$$

$\frac{1}{s+\frac{R}{L}} \cdot \frac{1}{L}$

상태천이 방정식 State-transition equation

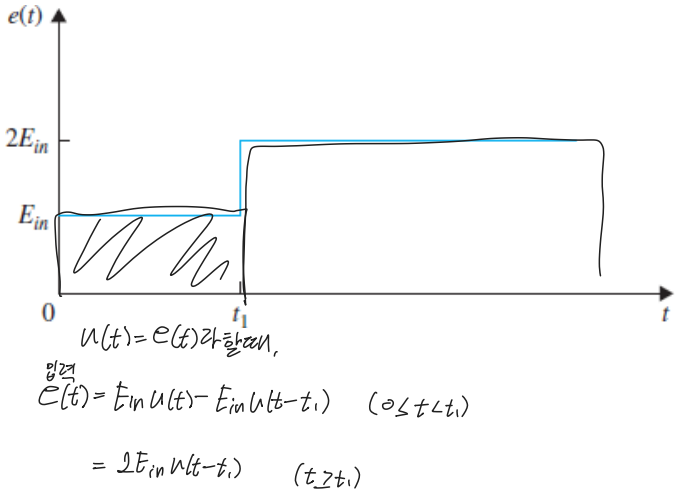


$$\frac{di(t)}{dt} = -\frac{R}{L}i(t) + \frac{1}{L}e_{in}(t)$$

$$A = -\frac{R}{L} \quad B = \frac{1}{L} \quad E = 0$$

$$\phi(t) = e^{-At} = e^{-Rt/L}$$

$$e(t) = E_{in}u_s(t) \quad 0 \le t < t_1$$



$$(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) = \frac{E_{in}}{Ls(s+R/L)} = \frac{E_{in}}{Rs[1+(L/R)s]}$$

$$i(t) = \left[e^{-Rt/L}i(0) + \frac{E_{in}}{R}(1-e^{-Rt/L})\right]u_s(t) \qquad 0 \le t < t_1$$

$$i(t_1) = e^{-Rt_1/L}i(0) + \frac{E_{in}}{R}(1-e^{-Rt_1/L})$$

$$i(t) = e^{-R(t-t_1)/L}i(t_1) + \frac{2E_{in}}{R}(1-e^{-R(t-t_1)/L}) \quad t \ge t_1$$

상태방정식과 고차 미분방정식의 관계

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \qquad y(t) = \mathbf{C}\mathbf{x}(t) = x_1(t)$$

$$\mathbf{C} = [1 \quad 0 \quad 0 \quad \cdots \quad 0]$$

$$\frac{dx_1(t)}{dt} = x_2(t)$$
$$\frac{dx_2(t)}{dt} = x_3(t)$$
$$\vdots$$
$$\frac{dx_n(t)}{dt} = -a_0 x_1(t) - a_1 x_2(t) - \cdots - a_{n-2} x_{n-1}(t) - a_{n-1} x_n(t) + f(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} (n \times n) \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} (n \times 1)$$

$$\frac{d^3y(t)}{dt^3} + 5\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + 2y(t) = u(t)$$

$$\frac{d^3y(t)}{dt^3} = -5\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) + u(t)$$

$$x_1(t) = y(t)$$
$$x_2(t) = \frac{dy(t)}{dt}$$
$$x_3(t) = \frac{d^2y(t)}{dt^2}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -1 & -5 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$y(t) = x_1(t) = [1 \quad 0]\mathbf{x}(t)$$

상태방정식과 전달함수 사이의 상호관계

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{E}\mathbf{w}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) + \mathbf{H}\mathbf{w}(t)$$

$\mathbf{x}(t) = n \times 1$ state vector
$\mathbf{u}(t) = p \times 1$ input vector
$\mathbf{y}(t) = q \times 1$ output vector
$\mathbf{w}(t) = v \times 1$ disturbance vector

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)]$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s) + \mathbf{H}\mathbf{W}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + \mathbf{C}(s\mathbf{I} - \mathbf{A})[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)] + \mathbf{D}\mathbf{U}(s) + \mathbf{H}\mathbf{W}(s)$$

전달함수의 정의는 초기조건이 0

$$\mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})\mathbf{E} + \mathbf{H}]\mathbf{W}(s)$$

G_u G_w

$$\mathbf{G}_u(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

$$\mathbf{G}_w(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{E} + \mathbf{H}$$

$$\mathbf{Y}(s) = \mathbf{G}_u(s)\mathbf{U}(s) + \mathbf{G}_w(s)\mathbf{W}(s)$$

CHANGWON NATIONAL UNIVERSITY

상태방정식과 전달함수 사이의 상호관계

$$\begin{cases} \dfrac{d^2y_1(t)}{dt^2}+4\dfrac{dy_1(t)}{dt}-3y_2(t)=u_1(t)+2w(t) \\ \dfrac{dy_1(t)}{dt}+\dfrac{dy_2(t)}{dt}+y_1(t)+2y_2(t)=u_2(t) \end{cases} \qquad \begin{cases} x_1(t)=y_1(t) \\ x_2(t)=\dfrac{dy_1(t)}{dt} \\ x_3(t)=y_2(t) \end{cases}$$

$$\begin{cases} \dot{x}_2+4x_2-3x_3=u_1+2w \\ \dot{x}_2+\dot{x}_3+x_1+2x_3=u_2 \end{cases}$$

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \\ \dfrac{dx_3(t)}{dt} \end{bmatrix}=\begin{bmatrix} 0 & 1 & 0 \\ 0 & -4 & 3 \\ -1 & -1 & -2 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}+\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}+\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}w(t)$$

A, B

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}=\mathbf{Cx}(t)$$

C

$$(s\mathbf{I}-\mathbf{A})=\begin{bmatrix} s & -1 & 0 \\ 0 & s+4 & -3 \\ 1 & 1 & s+2 \end{bmatrix}$$

$$=s\left((s+4)(s+2)+3\right)+3$$

$$|s\mathbf{I}-\mathbf{A}|=s^3+6s^2+11s+3$$

$$(s\mathbf{I}-\mathbf{A})^{-1}=\frac{1}{|s\mathbf{I}-\mathbf{A}|}\begin{bmatrix} s^2+6s+11 & s+2 & 3 \\ -3 & s(s+2) & 3s \\ -(s+4) & -(s+1) & s(s+4) \end{bmatrix}$$

$$\mathbf{G}_u(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}=\frac{1}{s^3+6s^2+11s+3}\begin{bmatrix} s+2 & 3 \\ -(s+1) & s(s+4) \end{bmatrix}$$

$$\mathbf{G}_w(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{E}=\frac{1}{s^3+6s+11s+3}\begin{bmatrix} 2(s+2) \\ -2(s+1) \end{bmatrix}$$

상태방정식과 전달함수 사이의 상호관계

$$\frac{d^2y_1(t)}{dt^2}+4\frac{dy_1(t)}{dt}-3y_2(t)=u_1(t)+2w(t)$$

$$\frac{dy_1(t)}{dt}+\frac{dy_2(t)}{dt}+y_1(t)+2y_2(t)=u_2(t)$$

$$\begin{bmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{bmatrix}\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix}=\begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}+\begin{bmatrix} 2 \\ 0 \end{bmatrix}W(s)$$

$$\mathbf{Y}(s)=\mathbf{G}_u(s)\mathbf{U}(s)+\mathbf{G}_w(s)\mathbf{W}(s)$$

$$\mathbf{G}_u(s)=\left[\begin{pmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{pmatrix}\right]^{-1}$$

$$\mathbf{G}_w(s)=\begin{bmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{bmatrix}^{-1}\begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\mathbf{G}_u(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}=\frac{1}{s^3+6s^2+11s+3}\begin{bmatrix} s+2 & 3 \\ -(s+1) & s(s+4) \end{bmatrix}$$

$$\mathbf{G}_w(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{E}=\frac{1}{s^3+6s+11s+3}\begin{bmatrix} 2(s+2) \\ -2(s+1) \end{bmatrix}$$

CHANGWON NATIONAL UNIVERSITY

특성방정식, 고유값과 고유벡터

✓특성방정식 Characteristic equations

$$\frac{d^n y(t)}{dt^n}+a_{n-1}\frac{d^{n-1}y(t)}{dt^{n-1}}+\cdots+a_1\frac{dy(t)}{dt}+a_0 y(t)$$
$$=b_m\frac{d^m u(t)}{dt^m}+b_{m-1}\frac{d^{m-1}u(t)}{dt^{m-1}}+\cdots+b_1\frac{du(t)}{dt}+b_0 u(t)$$

$$\left(s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0\right)y(t)=\left(b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0\right)u(t)$$

$$s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0=0$$

$$G(s)=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$$

$$\mathbf{G}_u(s)=\mathbf{C}\frac{\text{adj}(s\mathbf{I}-\mathbf{A})}{(s\mathbf{I}-\mathbf{A})}\mathbf{B}+\mathbf{D}$$
$$=\frac{\mathbf{C}[\text{adj}(s\mathbf{I}-\mathbf{A})]\mathbf{B}+|s\mathbf{I}-\mathbf{A}|\mathbf{D}}{|s\mathbf{I}-\mathbf{A}|}$$

$$|s\mathbf{I}-\mathbf{A}|=0$$

특성방정식, 고유값과 고유벡터

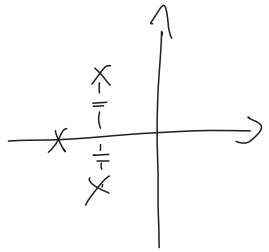
✓고유값 Eigenvalues

$$\frac{d^3y(t)}{dt^3}+5\frac{d^2y(t)}{dt^2}+\frac{dy(t)}{dt}+2y(t)=u(t)$$

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

$$\mathbf{A}=\begin{bmatrix}0&1&0\\0&0&1\\-2&-1&-5\end{bmatrix}\quad\mathbf{B}=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

원점의 좌 → stable한 시스템.



$$|s\mathbf{I}-\mathbf{A}|=\begin{vmatrix}s&-1&0\\0&s&-1\\2&1&s+5\end{vmatrix}=s^3+5s^2+s+2=0$$

$$s=-0.06047+j0.63738\quad s=-0.06047-j0.63738\quad s=-4.87906$$

CHANGWON NATIONAL UNIVERSITY

특성방정식, 고유값과 고유벡터

✓고유벡터 Eigenvectors

- 행렬방정식을 만족시키는 임의의 0이 아닌 벡터 $\mathbf{P}_i$

$$(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{p}_i = 0$$

$$\mathbf{P}_1 = \begin{bmatrix} p_{11} \\ p_{21} \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} p_{12} \\ p_{22} \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$|s\mathbf{I} - \mathbf{A}| = s^2 - 1$$

$AX = \lambda X$

$\begin{bmatrix} \lambda-1 & 1 \\ 0 & \lambda+1 \end{bmatrix}$

$\rightarrow (\lambda-1)(\lambda+1) = 0$

$\lambda^2 - 1 = 0$

$\therefore \lambda = \pm 1$

1) $\lambda = 1$

$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} P_{11} \\ P_{21} \end{bmatrix} = \begin{pmatrix} P_{21} = 0 \\ 2P_{21} = 0 \end{pmatrix} \Rightarrow P_{21} = 0$

P_{11}은 어떤 값이어도 성립 → 보통 가장 작은 값인 1을 대입.

2) $\lambda = -1$

$\begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} P_{12} \\ P_{22} \end{bmatrix} \rightarrow -2P_{12} + P_{22} = 0$

두 값의 비는 알 수 있다.

특성방정식, 고유값과 고유벡터

✓일반화된 Generalized Eigenvectors

- 행렬 A가 다중차수의 고유값을 가진다면 앞의 식을 이용하여 고유벡터를 구할 수 없다.
- 서로 다른 고유값을 가지면

$$(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{p}_i = 0$$

- 나머지 고유값은 아래 식으로 구할 수 있다.

$$(\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+2} = -\mathbf{p}_{n-q+1}$$
$$(\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+3} = -\mathbf{p}_{n-q+2}$$
$$\vdots$$
$$(\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+m} = -\mathbf{p}_{n-q+m-1}$$

특성방정식, 고유값과 고유벡터

$$(\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+1} = \mathbf{0}$$
$$(\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+2} = -\mathbf{p}_{n-q+1}$$

✓일반화된 Generalized Eigenvectors

$$\mathbf{A} = \begin{bmatrix} 0 & 6 & -5 \\ 1 & 0 & 2 \\ 3 & 2 & 4 \end{bmatrix} \qquad \lambda_1 = 2, \lambda_2 = \lambda_3 = 1$$

$$(\lambda_1 \mathbf{I} - \mathbf{A})\mathbf{p}_1 = \begin{bmatrix} 2 & -6 & 5 \\ -1 & 2 & -2 \\ -3 & -2 & -2 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{21} \\ p_{31} \end{bmatrix} = 0$$

$$\mathbf{p}_1 = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

$$(\lambda_2 \mathbf{I} - \mathbf{A})\mathbf{p}_2 = \begin{bmatrix} 1 & -6 & 5 \\ -1 & 1 & -2 \\ -3 & -2 & -3 \end{bmatrix} \begin{bmatrix} p_{12} \\ p_{22} \\ p_{32} \end{bmatrix} = 0 \qquad \mathbf{p}_2 = \begin{bmatrix} 1 \\ -\frac{3}{7} \\ -\frac{5}{7} \end{bmatrix}$$

$$(\lambda_3 \mathbf{I} - \mathbf{A})\mathbf{p}_3 = \begin{bmatrix} 1 & -6 & -5 \\ -1 & 1 & -2 \\ -3 & -2 & -3 \end{bmatrix} \begin{bmatrix} p_{13} \\ p_{23} \\ p_{33} \end{bmatrix} = -\mathbf{p}_2 = \begin{bmatrix} -1 \\ \frac{3}{7} \\ \frac{5}{7} \end{bmatrix}$$

$$\mathbf{p}_3 = \begin{bmatrix} 1 \\ -\frac{22}{49} \\ -\frac{46}{49} \end{bmatrix}$$

유사변환 Similarity transformation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

$$\mathbf{y}(\mathbf{t}) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t)$$

$$\mathbf{x}(t) = \mathbf{P}\bar{\mathbf{x}}(t)$$

$$\bar{\mathbf{x}}(t) = \mathbf{P}^{-1}\mathbf{x}(t)$$

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \bar{\mathbf{A}}\bar{\mathbf{x}}(t) + \bar{\mathbf{B}}u(t)$$

$$\bar{y}(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t) + \bar{\mathbf{D}}u(t)$$

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \mathbf{P}^{-1}\frac{d\mathbf{x}(t)}{dt} = \mathbf{P}^{-1}\mathbf{A}\mathbf{x}(t) + \mathbf{P}^{-1}\mathbf{B}u(t)$$
$$= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\bar{\mathbf{x}} + P^{-1}\mathbf{B}u(t)$$

$$\bar{y}(t) = \mathbf{C}\mathbf{P}\mathbf{x}(t) + \bar{\mathbf{D}}u(t)$$

$$\bar{\mathbf{C}} = \mathbf{C}\mathbf{P} \quad \bar{\mathbf{D}} = \mathbf{D}$$

유사변환 Similarity transformation

✓유사변환의 불변성

- 특성방정식, 고유값, 고유벡터와 전달함수가 이 변환으로 변하지 않는다.

✓특성방정식, 고유값과 고유벡터

$$\left|s\mathbf{I}-\bar{\mathbf{A}}\right|=\left|s\mathbf{I}-\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\right|=\left|s\mathbf{P}^{-1}\mathbf{P}-\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\right|$$

$$\left|s\mathbf{I}-\bar{\mathbf{A}}\right|=\left|\mathbf{P}^{-1}\right|\left|s\mathbf{I}-\mathbf{A}\right|\left|\mathbf{P}\right|=\left|s\mathbf{I}-\mathbf{A}\right|$$

✓전달함수

$$\begin{aligned}\bar{\mathbf{G}}(s)&=\bar{\mathbf{C}}(s\mathbf{I}-\bar{\mathbf{A}})\bar{\mathbf{B}}+\bar{\mathbf{D}}\\&=\mathbf{C}\mathbf{P}(s\mathbf{I}-\mathbf{P}^{-1}\mathbf{A}\mathbf{P})\mathbf{P}^{-1}\mathbf{B}+\mathbf{D}\end{aligned}$$

$$\bar{\mathbf{G}}(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})\mathbf{B}+\mathbf{D}=\mathbf{G}(s)$$

CHANGWON NATIONAL UNIVERSITY

유사변환 Similarity transformation

✓가제어성 표준형 Controllability canonical form

$$|s\mathbf{I}-\mathbf{A}| = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

$$\mathbf{P} = \mathbf{SM}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \ldots \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} a_1 & a_2 & \cdots & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & 1 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ a_{n-1} & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

$$\bar{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{AP} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

$$\bar{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

유사변환 Similarity transformation

✓가제어성 표준형 Controllability canonical form

$$\mathbf{A}=\begin{bmatrix}1&2&1\\0&1&3\\1&1&1\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}1\\0\\1\end{bmatrix}$$

$$|s\mathbf{I}-\mathbf{A}|=\begin{vmatrix}s-1&-2&-1\\0&s-1&-3\\-1&-1&s-1\end{vmatrix}=s^3-3s^2-s-3=0$$

$$\mathbf{M}=\begin{bmatrix}a_1&a_2&1\\a_2&1&0\\1&0&0\end{bmatrix}=\begin{bmatrix}-1&-3&1\\-3&1&0\\1&0&0\end{bmatrix}$$

$$\mathbf{S}=\begin{bmatrix}\mathbf{B}&\mathbf{AB}&\mathbf{A}^2\mathbf{B}\end{bmatrix}=\begin{bmatrix}1&2&10\\0&3&9\\1&2&7\end{bmatrix}$$

$$\mathbf{P}=\mathbf{SM}=\begin{bmatrix}3&-1&1\\0&3&0\\0&-1&1\end{bmatrix}$$

$$\bar{\mathbf{A}}=\mathbf{P}^{-1}\mathbf{AP}=\begin{bmatrix}0&1&0\\0&0&1\\3&1&3\end{bmatrix}$$

$$\bar{\mathbf{B}}=\mathbf{P}^{-1}\mathbf{B}=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

유사변환 Similarity transformation

✓가관측성 표준형 Observability canonical form

$$\mathbf{x}(t) = \mathbf{Q}\bar{\mathbf{x}}(t)$$

$$\mathbf{Q} = (\mathbf{MV})^{-1}$$

$$\bar{\mathbf{A}} = \mathbf{Q}^{-1}\mathbf{AQ} \quad \bar{\mathbf{B}} = \mathbf{Q}^{-1}\mathbf{B} \quad \bar{\mathbf{C}} = \mathbf{CQ} \quad \bar{\mathbf{D}} = \mathbf{D}$$

$$\bar{\mathbf{A}} = \mathbf{Q}^{-1}\mathbf{AQ} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}$$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} (n \times n)$$

$$\bar{\mathbf{C}} = \mathbf{CQ} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

유사변환 Similarity transformation

✓가관측성 표준형 Observability canonical form

$$\mathbf{A}=\begin{bmatrix}1&2&1\\0&1&3\\1&1&1\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}1\\0\\1\end{bmatrix}\quad \mathbf{C}=\begin{bmatrix}1&1&0\end{bmatrix}\quad \mathbf{D}=0$$

$$\bar{\mathbf{A}}=\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}=\begin{bmatrix}0&0&3\\1&0&1\\0&1&3\end{bmatrix}\quad \bar{\mathbf{C}}=\mathbf{C}\mathbf{Q}=\begin{bmatrix}0&0&1\end{bmatrix}\quad \bar{\mathbf{B}}=\mathbf{Q}^{-1}\mathbf{B}=\begin{bmatrix}3\\2\\1\end{bmatrix}$$

$$\mathbf{V}=\begin{bmatrix}\mathbf{C}\\\mathbf{CA}\\\mathbf{CA}^2\end{bmatrix}=\begin{bmatrix}1&1&0\\1&3&4\\5&9&14\end{bmatrix}$$

$$\mathbf{Q}=(\mathbf{MV})^{-1}=\begin{bmatrix}0.3333&-0.1667&0.3333\\-0.3333&0.1667&0.6667\\0.1667&0.1667&0.1667\end{bmatrix}$$

유사변환 Similarity transformation

✓대각표준형 Diagonal canonical form

$$\mathbf{x}(t) = \mathbf{T}\bar{\mathbf{x}}(t)$$

$$\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} \quad \bar{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B} \quad \bar{\mathbf{C}} = \mathbf{C}\mathbf{T} \quad \bar{\mathbf{D}} = \mathbf{D}$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} (n \times n)$$

$$\mathbf{T} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 & \cdots & \mathbf{p}_n \end{bmatrix}$$

$$\lambda_i \mathbf{p}_i = \mathbf{A}\mathbf{p}_i$$

$$\begin{bmatrix} \lambda_1\mathbf{p}_1 & \lambda_2\mathbf{p}_2 & \cdots & \lambda_n\mathbf{p}_n \end{bmatrix} = \begin{bmatrix} \mathbf{A}\mathbf{p}_1 & \mathbf{A}\mathbf{p}_2 & \cdots & \mathbf{A}\mathbf{p}_n \end{bmatrix} = \mathbf{A}\begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \end{bmatrix}\bar{\mathbf{A}} = \mathbf{A}\begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \end{bmatrix}$$

If the matrix **A** is of the CCF and **A** has distinct eigenvalues

$$\mathbf{T} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \lambda_3^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$$

유사변환 Similarity transformation

✓대각표준형 Diagonal canonical form

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

$\lambda_1 = -1$, $\lambda_2 = -2$, and $\lambda_3 = -3$

$$\mathbf{T} = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

유사변환 Similarity transformation

✓대각표준형 Diagonal canonical form

- 차원 축소 및 모델 저차원화

$$\mathbf{A} = \begin{bmatrix} -11 & -9 & 9 \\ 18 & 16 & -18 \\ 9 & 9 & -11 \end{bmatrix}$$

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \bar{\mathbf{A}}\bar{\mathbf{x}}(t) + \bar{\mathbf{B}}u(t)$$

$$\bar{y}(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t) + \bar{\mathbf{D}}u(t)$$

$$\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & -100 \end{bmatrix}$$

유사변환 Similarity transformation

✓Jordan 표준형 Jordan canonical form

$$\bar{\mathbf{A}} = \begin{bmatrix} \lambda_1 & 1 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & 0 & \lambda_2 & 0 \\ 0 & 0 & 0 & 0 & \lambda_3 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 6 & -5 \\ 1 & 0 & 2 \\ 3 & 2 & 4 \end{bmatrix} \qquad \lambda_1 = 2, \lambda_2 = \lambda_3 = 1$$

$$\mathbf{T} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ -1 & -\frac{3}{7} & -\frac{22}{49} \\ -2 & -\frac{5}{7} & -\frac{46}{49} \end{bmatrix}$$

$$\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$